

1 Base Drag Dominates the Apogee Error Budget in
2 Open-Source Supersonic Rocket Simulation: A
3 Mechanism-Attribution Study with External
4 Base-Pressure Benchmarks

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8 **Abstract**

9 Base drag is the least-constrained term in slender-body supersonic aerody-
10 namics, yet its weight in an *integrated* apogee prediction is rarely isolated.
11 We ask how heavily the base-drag term weighs on the apogee of fin-stabilized
12 supersonic sounding rockets relative to the other supersonic submodels, and
13 whether the closure that carries that weight can be anchored to external base-
14 pressure data. The base-drag closure as implemented—turbulent Chapman–
15 Korst shear-layer theory, a Chapman laminar correlation, an ESDU 77021
16 boundary-layer-thickness correction, an empirical power-law base-drag cor-
17 relation, a Viswanath boattail correction, and a finned-base augmentation
18 term—is benchmarked against external, out-of-sample base-pressure mea-
19 surements, and a controlled mechanism ablation then disables each drag
20 contribution in isolation across an archived 24-flight subset (23 single-stage
21 flights plus the AeroPac 104K two-stage closure) spanning Mach 0.54 to 3.46.
22 The turbulent closure reproduces the reference NACA TN 3393 nonlifting-

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body base pressures to a mean absolute percentage error of 15.9% (and the Hart free-flight subset to 4.0%), and the Chapman laminar branch to 4.4%. The ablation shows that base-drag augmentation moves the mean apogee error by 8.10 percentage points—the dominant mechanism, roughly nine times the next supersonic submodel (compressible skin friction, 0.87 percentage points) and about 54 times the shock-geometry pre-pass (0.15 percentage points). A decontaminated prospective holdout, on which the model is more accurate than on the development flights (mean absolute error 3.95% versus 5.47%), shows that the corpus-frozen base-drag constants generalize rather than overfit. We frame these results as a mechanism-attribution study, backed by external base-pressure benchmarks, that locates base drag at the top of the apogee error budget—not a per-closure intercomparison, and not an integrated-validation or parity claim—complementing the integrated study reported separately.

Keywords: base drag, boattail, supersonic rocket, trajectory simulation, mechanism ablation, Chapman–Korst, apogee prediction

1. Introduction

For a slender finned rocket flying supersonically, the pressure deficit acting over the blunt aft face — the base drag — is not a small correction. Once the boundary layer separates at the base and the recirculating wake closes behind a free shear layer, the base pressure falls well below ambient, and the resulting axial force is a large fraction of the total zero-lift drag: of

45 order one quarter to one half across the low-supersonic band where most
 46 amateur and sounding-rocket flights spend the bulk of their powered and
 47 coasting time. The contribution peaks near $M \approx 1.05$, where the base-drag
 48 coefficient of a flat-based body reaches roughly $C_{d,\text{base}} \approx 0.25$, and then
 49 decays as the freestream entrains the wake at higher Mach number. Because
 50 this term is both large and concentrated in the speed range that dominates
 51 the climb-to-apogee integral, the choice of base-drag closure is one of the
 52 most consequential modeling decisions in a supersonic trajectory code.

53 It is also one of the least settled. Base pressure resists first-principles
 54 prediction — it depends on boundary-layer state at separation, base geome-
 55 try, afterbody taper, and, for a finned vehicle, on the fins themselves — so
 56 the engineering literature offers a proliferation of competing closures rather
 57 than a single accepted law. Chapman’s free-shear-layer analysis and the
 58 Chapman–Korst recompression model underpin the turbulent base-pressure
 59 estimates that remain the reference for nonlifting bodies of revolution Reller
 60 and Hamaker (1955); the ESDU 77021 data item consolidates a broad exper-
 61 imental corpus into a recommended base-pressure correlation for supersonic
 62 and hypersonic Mach numbers Engineering Sciences Data Unit (1977); and
 63 practical rocketry codes often fall back on compact empirical fits such as
 64 $C_{d,\text{base}} = 0.064 + 0.186/M^2$, which captures the observed high-Mach decay
 65 with two coefficients. Layered on top of these are geometry-specific modifiers:
 66 a boattail reduces base area and turns the flow, cutting base drag by a sub-
 67 stantial margin for well-chosen aft-taper angles Viswanath (1996), whereas a

68 blunt base does not; and a four-fin tail can either relieve or augment the base
69 suction relative to the clean-body value, depending on how the fin wakes and
70 the body wake interact. The closures disagree not only in magnitude but
71 in functional form, and each was calibrated on a particular geometry fam-
72 ily. High-fidelity computation has since resolved the near-wake structure in
73 detail — thin-layer Navier–Stokes solutions recover the transonic base flow
74 of an ogive-cylinder-boattail projectile, for instance Sahu et al. (1983) —
75 but such methods are too costly to embed inside a six-degree-of-freedom tra-
76 jectory integrator that must evaluate aerodynamic coefficients thousands of
77 times per flight, so engineering codes still rely on the semi-empirical closures
78 above. The open question this paper addresses is therefore not which closure
79 is most physically complete, but *how much* the base-drag term as a whole
80 weighs on the single number a designer cares about — the predicted apogee
81 — relative to the other supersonic submodels, and whether the closure that
82 carries that weight can be anchored to external base-pressure measurements
83 rather than to the flight corpus alone.

84 Despite the maturity of the individual closures, the open literature does
85 not quantify, in a reproducible way, how heavily the base-drag term weighs
86 in a full six-degree-of-freedom flight against the other supersonic submodels
87 — the single number a designer ultimately cares about being the predicted
88 apogee. Component-level validations report base-pressure or base-drag error
89 against wind-tunnel data, but a base-drag coefficient that matches a tunnel
90 point to within a few percent may still move apogee by far more, or far less,

91 than its component error suggests, because apogee integrates the drag his-
 92 tory over a trajectory weighted toward the lower-Mach portion of the flight.
 93 Conversely, a submodel with a larger component error can be nearly inert
 94 at apogee if its disagreement falls outside the speed range that dominates
 95 the integral. The established open-source simulator OpenRocket Niskanen
 96 (2009) did not model supersonic base drag at all, while the commercial ref-
 97 erence RASAero II Rogers (2015) embeds its base-drag treatment in closed
 98 source, so neither tool exposes the per-mechanism apogee sensitivity in a
 99 form that can be audited or reproduced.

100 The companion study Yu (2026a) established the integrated context for
 101 the present work: an open-source supersonic extension of OpenRocket, val-
 102 idated against a 25-flight external corpus spanning Mach 0.54 to 4.33, that
 103 achieves apogee-prediction parity with the commercial RASAero II baseline.
 104 A mechanism ablation reported in that work isolated the contribution of
 105 each supersonic model to the corpus apogee error and produced a striking
 106 ranking. Disabling the finned-base augmentation shifted mean apogee error
 107 by 8.10 percentage points (up to 39.5 percentage points on the Mach-2.19
 108 A-601 Kinsel flight) — by far the dominant driver of integrated apogee ac-
 109 curacy. The remaining mechanisms were comparatively minor: compress-
 110 ible (Van Driest II) skin friction accounted for 0.87 percentage points, the
 111 DATCOM 4.1.5.1 fin wave-drag term for 0.39 percentage points, and the
 112 shock-geometry pre-pass — the celebrated architectural innovation of the
 113 companion paper — for only 0.15 percentage points, because that pre-pass is

114 inert below Mach 1 and apogee integrates a trajectory dominated by lower-
115 Mach drag. In other words, the base-drag closure, not the shock geometry,
116 governs the apogee error budget.

117 This paper takes that ablation result as its point of departure and asks
118 a distinct research question: *how much does the base-drag term weigh in the*
119 *integrated supersonic-rocket apogee error budget relative to the other super-*
120 *sonic submodels, and can the closure that carries that weight be anchored*
121 *to external base-pressure measurements?* Where the companion paper Yu
122 (2026a) reports an integrated parity result and treats base drag as one com-
123 ponent among many, this paper diverges into a focused mechanism attri-
124 bution supported by external component benchmarks. The contribution
125 has three parts. First, we describe the base-drag closure as implemented
126 — the Chapman–Korst turbulent shear-layer model Reller and Hamaker
127 (1955), the Chapman laminar correlation Chapman (1950), the ESDU 77021
128 boundary-layer-thickness correction Engineering Sciences Data Unit (1977),
129 the empirical $C_{d,\text{base}} = 0.064 + 0.186/M^2$ correlation (presented as an em-
130 pirical fit validated against the NACA TN 3393 base-pressure data Reller
131 and Hamaker (1955) and consistent with ESDU 77021 Engineering Sciences
132 Data Unit (1977), with no reliance on any unverifiable correlation), the
133 Viswanath boattail correction Viswanath (1996), and a finned-base augmen-
134 tation term — and we *benchmark* its branches against external, out-of-sample
135 base-pressure data (NACA TN 3393 turbulent and laminar, and the Hart
136 NACA RM L52E06 free-flight set). Alternative literature closures exist for

several of these branches, and we note where they differ in form and calibration geometry; we do *not*, however, substitute them one at a time and report per-closure apogee deltas — that closure-swap experiment is outside the scope of this study. Second, we report a controlled *mechanism ablation* that disables each implemented aerodynamic mechanism in isolation across the corpus and quantifies how much each moves integrated apogee, localizing the base-drag term at the top of the error budget. Third, we confront the in-sample exposure of the corpus-frozen base-drag constants head-on with a decontaminated prospective holdout, which tests whether they generalize. The remainder of the paper specifies the closure and its components, details the benchmark and ablation methodology, reports the component-benchmark, ablation, and holdout results, and discusses what the ranking implies for open supersonic trajectory modeling.

2. The Supersonic Base-Drag Closure and Its Components

This section specifies the supersonic base-drag closure as implemented and its component branches. Each branch contributes to the base-drag coefficient $C_{d,\text{base}}$ referenced to the base area $A_{\text{base}} = \pi R_{\text{base}}^2$, so that its contribution to the vehicle axial-force coefficient is $C_{d,\text{base}} A_{\text{base}}/A_{\text{ref}}$. For each branch we state its governing equation, its regime of validity, and the published source against which it was assessed. Three of the six components are geometry-independent functions of Mach number and boundary-layer state — the turbulent Chapman–Korst shear-layer model, the Chapman laminar

correlation, and the empirical supersonic correlation — consolidated under the ESDU 77021 data item; the remaining three are geometry modifiers applied multiplicatively to the clean-body value: the ESDU 77021 boundary-layer-thickness correction, the Viswanath boattail relief, and the finned-base augmentation. The branches are stitched together below Mach 1.5 by a single C^1 -continuous transonic blend, described last. Table 1 collects the inventory. The engineering literature offers *alternative* closures for several of these branches — competing turbulent base-pressure correlations, different boattail-relief models, and several finned-base treatments — that disagree in functional form and were calibrated on different geometry families; we note these alternatives where relevant, but this study describes and benchmarks the closure as implemented rather than substituting alternatives one at a time and reporting per-closure apogee deltas, which is outside its scope.

2.1. Empirical supersonic correlation

The workhorse turbulent closure is the two-coefficient empirical correlation

$$C_{d,\text{base}} = 0.064 + \frac{0.186}{M^2}, \quad M > 1.3, \quad (1)$$

applicable to a slender axisymmetric afterbody with a turbulent boundary layer at separation. The two coefficients encode the two physical asymptotes of supersonic base flow: the leading constant 0.064 is the nonzero high-Mach floor toward which the base-pressure coefficient asymptotes as the freestream entrains the wake, and the $0.186/M^2$ term captures the decay

180 of the base-pressure deficit with dynamic pressure as Mach number rises.
 181 We present Eq. (1) strictly as an empirical fit: it reproduces the turbulent
 182 NACA TN 3393 Reller and Hamaker (1955) nonlifting-body base-pressure
 183 measurements over $M = 2.73$ – 4.48 to a mean absolute percentage error of
 184 15.9% (and the Hart free-flight subset to 4.0%), and its functional form is
 185 consistent with the recommended supersonic base-pressure correlation con-
 186 solidated in ESDU 77021 Engineering Sciences Data Unit (1977). No further
 187 provenance is claimed for the two coefficients beyond this benchmark agree-
 188 ment.

189 *2.2. Chapman laminar base drag*

190 For perfect-finish vehicles whose surface is smooth enough that boundary-
 191 layer transition is delayed and the layer remains laminar to the base, the
 192 turbulent form of Eq. (1) over-predicts the base suction. The Chapman
 193 laminar correlation Chapman (1950) applies instead:

$$C_{p,b,\text{lam}} = \frac{C_{\text{lam}}}{M^2 \sqrt{\text{Re}_L}}, \quad C_{\text{lam}} = 1870, \quad (2)$$

194 with Re_L the body-length Reynolds number. The $\text{Re}_L^{-1/2}$ scaling follows
 195 from laminar free-shear-layer mixing theory — a laminar layer carries less
 196 momentum, mixes less, and recompresses to a lower base pressure than a
 197 turbulent layer — and the M^{-2} factor again arises from normalizing the base-
 198 pressure deficit by freestream dynamic pressure. The constant $C_{\text{lam}} = 1870$
 199 is a geometric-mean fit to the four condensation-corrected laminar TN 3393

200 points ($M = 2.73\text{--}4.48$, $\text{Re}_L \approx 4\text{--}6 \times 10^6$), giving a mean absolute percentage
 201 error of 4.4% against those data. Applying the turbulent form of Eq. (1) to
 202 the same laminar data yields a mean absolute percentage error of 44%: the
 203 boundary-layer state at separation is the dominant discriminator between
 204 the two branches, which is precisely why a base-drag benchmark cannot treat
 205 “the” base-drag closure as a single object. Equation (2) is blended into the
 206 turbulent result over $M = 1.3\text{--}2.5$ by the Hermite smoothstep $s(t) = 3t^2 - 2t^3$,
 207 weighted by the local laminar fraction so that a fully turbulent body never
 208 sees the laminar branch; a clamp at the near-vacuum limit $2/(\gamma M^2)$ prevents
 209 unphysical growth at low Reynolds number.

210 *2.3. Chapman–Korst turbulent shear-layer model*

211 The Chapman–Korst dead-air recompression theory Reller and Hamaker
 212 (1955) provides a turbulent base-pressure estimate from boundary-layer edge
 213 conditions, used here as the supersonic-side closure that the empirical corre-
 214 lation blends *into*. The supersonic shear layer separating at the base corner
 215 reattaches on the wake axis at a recompression point; the balance between
 216 corner expansion and shear-layer recompression fixes the base pressure. For
 217 a thin turbulent boundary layer the clean-body coefficient follows the same
 218 family as Eq. (1) with an added M^{-4} term, calibrated to the turbulent
 219 cylindrical-afterbody tabulation of ESDU 77021 Engineering Sciences Data
 220 Unit (1977):

$$C_{d,\text{base}}^{\text{thin}} = 0.060 + \frac{0.190}{M^2} + \frac{0.005}{M^4}. \quad (3)$$

221 This reproduces the ESDU anchor values $C_{d,\text{base}} \approx 0.145, 0.110, 0.085$, and
 222 0.070 at $M = 1.5, 2.0, 3.0$, and 5.0 , respectively, which agree closely with
 223 Eq. (1) for a thin layer — as expected, since both describe the same recom-
 224 pression physics. Equation (3) is the baseline to which the boundary-layer-
 225 thickness correction of Section 2.4 is applied. It is blended into the empirical
 226 correlation over $M = 1.2$ – 1.4 by the same smoothstep $s(t)$, so that pure
 227 Eq. (1) governs below $M = 1.2$ and the Chapman–Korst form governs above
 228 $M = 1.4$.

229 *2.4. ESDU 77021 boundary-layer-thickness correction*

230 ESDU 77021 Engineering Sciences Data Unit (1977) consolidates a broad
 231 experimental corpus of base pressure on bodies of revolution at supersonic
 232 and hypersonic Mach numbers without base injection or combustion, and is
 233 the reference data item for the turbulent closures above. Beyond fixing the
 234 thin-layer baseline of Eq. (3), it supplies the dependence on boundary-layer
 235 thickness: a thicker layer entrains more mass into the recirculating wake,
 236 raises base pressure, and so *reduces* base drag. The correction is applied as
 237 a multiplicative factor on the thin-layer coefficient,

$$f_\delta = 1 - k(M)\sqrt{\theta/R_{\text{base}}}, \quad k(M) = 0.8 + \frac{0.2}{M}, \quad (4)$$

238 where θ is the turbulent momentum thickness at the base, estimated from the
 239 flat-plate correlation $\theta/x = 0.036 \text{Re}_x^{-1/5}$, and θ/R_{base} is clamped to $[10^{-5}, 0.5]$
 240 with f_δ clamped to $[0.3, 1.0]$. For a typical rocket ($\theta/R_{\text{base}} \approx 0.01$) the

reduction is a few percent; for an unusually thick layer ($\theta/R_{\text{base}} \approx 0.1$) it reaches 20–30%. This correction is the mechanism by which the slender, high-fineness airframes in the corpus carry a distinct base-drag signature from short, blunt bodies, and it is the literature-anchored counterpart to the corpus-frozen thick-boundary-layer amplification disclosed in the companion study Yu (2026a).

2.5. *Viswanath boattail relief*

When the afterbody tapers inward to a boattail, the converging flow turns through a compression that raises base pressure and shrinks the base area, reducing base drag below the blunt-base value. Following Viswanath Viswanath (1996), the relief is applied as a multiplicative retention factor η_{bt} on the clean-body base drag, parameterized by the boattail half-angle $\theta_{bt} = \arctan[(R_{\text{fore}} - R_{\text{aft}})/L_{bt}]$ and, in the effective regime, by Mach number:

$$\eta_{bt} = \begin{cases} 0.25 + 0.05 \theta_{bt}, & \theta_{bt} < 6^\circ, \\ [0.55 + 0.04(\theta_{bt} - 6^\circ)] [1 + 0.1 \max(0, M - 1)], & 6^\circ \leq \theta_{bt} < 16^\circ, \\ 0.95 - 0.05(\theta_{bt} - 16^\circ), & \theta_{bt} \geq 16^\circ, \end{cases} \quad (5)$$

with η_{bt} clamped to $[0, 1]$. For well-chosen boattail angles of 6–16° this yields a 15–40% base-drag reduction, consistent with the afterbody-drag-management survey of Viswanath Viswanath (1996). The factor is applied to the wake *downstream* of a boattail and is not also applied on the boattail

258 component itself, which would double-count the relief and can drive the aft
 259 base drag to zero on steep imported geometries that still expose a finite blunt
 260 exit area.

261 2.6. Finned-base augmentation (externally calibrated, B-level)

262 The dominant apogee mechanism (§4.2) is the augmentation of base drag
 263 by the tail fins. Fins mounted at or just ahead of the base disrupt the re-
 264 compression of the body wake, deepening the base suction relative to the
 265 clean-body value; aeroballistic-range data on the Basic Finner Dupuis and
 266 Hathaway (1997) and the afterbody-drag compendium of Hoerner Hoerner
 267 (1965) consistently show 40–60% higher base drag for four-fin configurations
 268 than for the corresponding smooth cylindrical afterbody over $M = 1.5$ –3,
 269 which we use as an external *bound* on the plausible magnitude of the ef-
 270 fect rather than as a calibration target. The augmentation is applied as
 271 a multiplicative factor on the post-relief base drag, written here in its full
 272 implemented form,

$$\Phi_{\text{fin}} = 1 + K_{\text{fin}} \frac{1 - e^{-n_{\text{fins}}/1.4}}{1 - e^{-4/1.4}} \text{clamp}\left(\frac{s_{\text{max}}}{R_{\text{base}}}, 0.3, 1\right) f(M), \quad K_{\text{fin}} = 0.55, \quad (6)$$

273 where the fin count n_{fins} enters through a saturating term normalized to
 274 unity at the four-fin reference (wake disruption is not linear in fin count:
 275 three corner-wake sectors already nearly divide the base shear layer, so a
 276 three-fin body is not under-counted), s_{max} is the maximum fin span, and
 277 the Mach ramp $f(M)$ rises from 0 at $M = 0.2$ to 0.30 at $M = 0.8$, then

278 from 0.30 to 1 over $M = 0.8$ – 1.3 , holds through the low-supersonic band to
 279 $M = 3.0$, then decays as $3/M$ above $M = 3.0$ as the fins become progressively
 280 submerged in the body shock cone. At the four-fin reference (four rectangular
 281 fins, $s_{\max}/R_{\text{base}} = 2.0$, where both geometry factors saturate to unity) the
 282 scale constant $K_{\text{fin}} = 0.55$ gives a $\sim 50\%$ augmentation at $M = 2$, inside
 283 the 40–60% literature bound above; companion scale constants of 1.00 for
 284 rounded-trailing-edge fins and 1.35 for fin-can sleeve geometries cover the
 285 remaining corpus afterbody types.

286 We classify K_{fin} as a *B-level* constant in the specific sense that it lacks
 287 validation against an isolated finned-base *base-pressure* dataset. It is *not*
 288 corpus-frozen: the value 0.55 is calibrated externally against the Basic Finner
 289 aeroballistic-range data Dupuis and Hathaway (1997) so that it reproduces
 290 the $\sim 50\%$ augmentation measured there at $M = 2$, which also places it inside
 291 the 40–60% over-clean-body band of the same fixture and the Hoerner com-
 292 pendium Hoerner (1965). Its anchor is thus a published external fixture, not
 293 flights drawn from the calibration corpus, so it does not make the headline in-
 294 sample. The constant nonetheless does not yet carry the A-level external val-
 295 idation of Eqs. (1) and (2), which are checked against isolated base-pressure
 296 data, and this B-to-A gap is the explicit limitation of the augmentation clo-
 297 sure. The distinction is material precisely because the mechanism ablation of
 298 §4.2 ranks this term as the dominant driver of integrated apogee error, shift-
 299 ing mean corpus apogee error by 8.10 percentage points when removed —
 300 about 54 times the 0.15-point effect of the shock-geometry pre-pass. The two

301 genuinely corpus-frozen constants in the base-drag treatment belong to *different*
 302 *ent* closures — the thick-boundary-layer base-drag amplifier (THICK_BL_K)
 303 and the slender-body pressure-drag term (SLENDER_BODY_K) — and it is
 304 those two constants, not K_{fin} , that the decontaminated prospective holdout
 305 of §4.3 quarantines and shows to generalize (blind holdout MAE 3.95% below
 306 development 5.47%). The defense of K_{fin} is instead its external Basic Finner
 307 anchor and the component base-pressure benchmarks. Elevating K_{fin} from
 308 this total-drag anchor to A-level validation against an isolated finned-base
 309 base-pressure dataset is identified in the Conclusions as the single highest-
 310 value evidence gap raised by this work.

311 2.7. Transonic blend and C^1 continuity

312 Below $M = 1.5$ the supersonic closures above are not valid, and the
 313 base drag instead rises to a transonic peak near $M \approx 1.05$, where a flat-
 314 based body reaches $C_{d,\text{base}} \approx 0.25$. The transonic band $M = 0.85$ – 1.5 is
 315 represented by a degree-4 polynomial anchored at three points: the peak
 316 value $C_{d,\text{base}} = 0.25$ at $M = 1.05$, the free-flight plateau $C_{d,\text{base}} = 0.230$
 317 at $M = 1.30$ (just below the Hart NACA RM L52E06 Hart (1952) finless
 318 free-flight reading of 0.250 ± 0.013 , within its digitization uncertainty), and
 319 a C^1 handoff to the supersonic correlation at $M = 1.5$. The polynomial
 320 is constructed to be continuous in both value and first derivative at the
 321 regime boundaries, and the laminar, Chapman–Korst, and boattail blends
 322 of the preceding subsections all use the same Hermite smoothstep $s(t) =$

323 $3t^2 - 2t^3$, which has zero derivative at its endpoints. This global C^1 continuity
 324 is not cosmetic: a discontinuity in $C_{d,\text{base}}$ or its slope near Mach 1 drives
 325 the fourth-order Runge–Kutta trajectory integrator to oscillate whenever the
 326 vehicle repeatedly crosses the sonic line during boost and coast, and the
 327 resulting integration noise would contaminate the very apogee sensitivity
 328 this paper sets out to measure. The same continuity discipline is applied
 329 across every Mach-regime transition in the underlying simulator, as detailed
 330 in the companion paper Yu (2026a).

331 **3. Benchmark and Mechanism-Ablation Methodology**

332 The methodology has three parts, ordered from the most external evi-
 333 dence to the most in-sample. The *component intercomparison* (Section 3.1)
 334 is the substantive physics of the paper: it asks how each candidate base-
 335 drag closure reproduces *measured* base pressure in isolation, against external
 336 wind-tunnel and free-flight data that played no part in any corpus calibra-
 337 tion. The *mechanism ablation* (Section 3.2) is the motivating experiment: it
 338 asks how much each implemented aerodynamic mechanism actually moves
 339 integrated apogee, by disabling it one at a time across the flight corpus, and
 340 it is what singles out base drag as the dominant driver. The *decontaminated*
 341 *holdout* (Section 3.3) is the in-sample defense: it asks whether the two corpus-
 342 frozen base-drag scale constants generalize to flights that never informed their
 343 calibration, pre-empting the circularity critique that the ablation substrate
 344 would otherwise invite. We lead with the externally validated closure fidelity,

345 use the ablation as the finding that motivates a base-drag study at all, and
346 confront the in-sample constants head-on rather than letting them read as
347 the result.

348 3.1. Component intercomparison protocol

349 Each base-drag closure introduced in Section 2 is exercised against pub-
350 lished base-pressure data with the boundary-layer state and length Reynolds
351 number Re_L matched to the reference condition, so that closures are com-
352 pared on the flow regime each was derived for rather than averaged across
353 regimes. The discriminator is deliberate: as Section 2.2 shows, applying a
354 turbulent correlation to laminar separation data inflates the error by an order
355 of magnitude, so the protocol conditions closure selection on the boundary-
356 layer state and not on Mach number alone. The error metric throughout is
357 the mean absolute percentage error (MAPE) of the predicted base-pressure
358 or base-drag coefficient against the digitized reference points. Three exter-
359 nal datasets anchor the comparison, none of which was used to tune any
360 integrated-trajectory constant; a fourth, computational, comparator is iden-
361 tified but deferred.

362 *NACA TN 3393 (turbulent and laminar)*.. The primary anchor is the Reller
363 and Hamaker base-pressure investigation of nonlifting bodies of revolution at
364 Mach 2.73–4.98 Reller and Hamaker (1955), which reports base pressure for
365 both turbulent and laminar boundary-layer states on the same geometries
366 and therefore exercises both base-drag branches against a single, internally

367 consistent experiment. Against the turbulent points the empirical correlation
 368 $C_{d,\text{base}} = 0.064 + 0.186/M^2$ of Eq. (1) attains a MAPE of 15.9%; against the
 369 laminar points the Chapman laminar closure of Eq. (2) attains 4.4% Reller
 370 and Hamaker (1955). The laminar closure is thus markedly more faithful on
 371 its target regime, and the two branches are not interchangeable: applying
 372 the turbulent correlation to the same laminar data yields a MAPE of 44%.
 373 This is the quantitative basis for treating “the” base-drag closure as a state-
 374 dependent family rather than a single object.

375 *Hart NACA RM L52E06 (transonic turbulent)..* To extend the turbulent
 376 comparison below the TN 3393 floor and through the transonic base-drag
 377 peak near $M \approx 1.05$, the turbulent closure is additionally compared against
 378 the Hart free-flight base-pressure measurements (NACA RM L52E06 Hart
 379 (1952)), where it attains a MAPE of 4.0%. The two turbulent anchors to-
 380 gether span the transonic peak (Hart) and the supersonic decay (TN 3393),
 381 and bracket the band in which the empirical correlation $C_{d,\text{base}} = 0.064 +$
 382 $0.186/M^2$ and the degree-4 transonic blend of Section 2.7 operate. The Hart
 383 finless free-flight reading of 0.250 ± 0.013 near the peak also fixes the transonic
 384 anchor point of that blend.

385 *Chapman laminar data..* The laminar branch is anchored on the laminar
 386 subset of the same TN 3393 experiment Reller and Hamaker (1955), against
 387 which the Chapman laminar closure attains the 4.4% MAPE quoted above.
 388 This isolates the “perfect-finish” / delayed-transition case relevant to highly

389 polished slender vehicles, for which the turbulent correlation over-predicts
390 base drag, and provides the externally validated counterpart to the laminar
391 branch invoked for the smoothest airframes in the corpus.

392 *Transonic CFD comparator (deferred, disclosed)*.. A fourth comparison is
393 identified but not executed in this work: the Sahu, Nietubicz and Steger
394 thin-layer Navier–Stokes computation of base flow on a secant-ogive-cylinder-
395 boattail projectile at transonic speed ($M \approx 0.9$ – 1.2) Sahu et al. (1983) would
396 provide an independent, CFD-grade check of the closures through the tran-
397 sonic peak, where the empirical and Chapman–Korst forms hand off to the
398 degree-4 blend. Its geometry is structurally different from the projectile fam-
399 ilies used for the other anchors, so reproducing it would require building a
400 separate vehicle model and digitizing the published base-pressure distribu-
401 tions. We disclose this comparison as *deferred*—it is documented as a memo
402 only and is not exercised as a quantitative comparator here—rather than
403 report a partial result. We claim *no computational fluid dynamics of our*
404 *own*; Sahu et al. (1983) is a published third-party computation cited as a
405 prospective comparator.

406 3.2. *Controlled mechanism-ablation protocol*

407 The component intercomparison establishes closure fidelity in isolation
408 but does not establish which mechanism governs the *integrated* apogee error.
409 A closure that matches a wind-tunnel point to a few percent may still move
410 apogee by much more, or much less, than its component error suggests, be-

411 cause apogee integrates the drag history over a trajectory weighted toward
 412 the lower-Mach portion of the flight. To resolve this, we run a controlled
 413 ablation on the archived 24-flight mechanism-ablation subset of the exter-
 414 nally selected supersonic altitude-comparison corpus, spanning Mach 0.54–
 415 3.46, used for the integrated validation reported in the companion flagship
 416 study Yu (2026a). The ablation subset contains 23 single-stage flights plus
 417 the AeroPac 104K two-stage closure. MESOS 293K remains part of the com-
 418 panion 25-flight validation corpus and is reported there as the highest-Mach
 419 two-stage case; it is not part of the archived mechanism-ablation artifact
 420 from which the 8.10/0.87/0.39/0.15/0.00 pp ranking is computed. Here the
 421 corpus is the *substrate for a mechanism-attribution experiment*, not the basis
 422 for an accuracy headline.

423 The protocol disables one aerodynamic mechanism at a time, re-flies all
 424 24 of these trajectories, and records the change in the corpus mean absolute
 425 apogee error, $|\overline{\Delta}|$, relative to the fully enabled baseline. Five mechanisms
 426 are ablated: the finned-base drag augmentation Φ_{fin} of Eq. (6); the Van Dri-
 427 est II compressible skin-friction transformation Hopkins and Inouye (1971);
 428 the DATCOM Section 4.1.5.1 fin wave-drag closure Finck (1978); Ulmann
 429 (1956); the shock-geometry pre-pass; and the Pitts–Nielsen–Kaattari fin-
 430 body interference correction, together with its K_1 floor.

431 Each mechanism is ablated *in isolation*—disabled singly, with every other
 432 mechanism held at its baseline state, and the corpus re-run from a clean
 433 configuration for each ablation. This isolation discipline is essential and

not incidental: the ablation flags are evaluated as shared static state, so toggling several mechanisms within a single process risks cross-contaminating the recorded baseline and corrupting the attribution. Running each ablation as an independent, clean corpus pass removes that contamination, and the resulting single-mechanism apogee sensitivities are the per-mechanism $\overline{|\Delta|}$ values reported in Section 4.

The ranking that emerges—and the finding that motivates this paper—is steeply ordered. The finned-base augmentation dominates the apogee error budget at a mean $\overline{|\Delta|}$ of 8.10 pp (reaching 39.5 pp on the Mach-2.19 A-601 Kinsel flight), more than an order of magnitude larger than the next mechanism. Compressible skin friction contributes 0.87 pp, the DATCOM fin wave-drag term 0.39 pp, and the shock-geometry pre-pass—the celebrated architectural innovation of the companion paper—only 0.15 pp, because that pre-pass is inert below Mach 1 and apogee integrates a trajectory dominated by lower-Mach drag; the PNK interference term, with its K_1 floor, registers no measurable apogee effect (0.00 pp). Base drag, not shock geometry, governs the apogee error budget, which is what licenses a focused base-drag closure study distinct from the integrated parity result of Yu (2026a).

3.3. Decontaminated prospective holdout for the corpus-frozen constants

The turbulent base-drag branch carries two scale constants that were frozen against the corpus rather than against an independent component dataset: a thick-boundary-layer scale, $\text{THICK_BL_K} = 2.2$, and a slender-

body scale, $\text{SLENDER_BODY_K} = 0.0025$. Because these constants were tuned on corpus flights, the integrated result is *partly in-sample*, and a naive corpus-wide accuracy figure would be optimistically biased. We disclose this plainly and test generalization with a decontaminated prospective holdout, which is the defense against the circularity critique that follows from using a corpus-calibrated quantity as the ablation substrate of Section 3.2.

The decontamination rule is strict: *every* flight that either constant touched during calibration—the anchor flights for both constants and the source-diagnostic flight—is quarantined in the development partition. This decontamination is applied as an overlay on the corpus’s existing development/holdout bookkeeping: the five flights involved in fixing the two constants (Raven, Rabia, Rabia Short Fin Can, Kinsel, and Torrent) are placed in the development partition (DEV) together with the cases already reserved for closure development and ablation, while no calibration-touched flight is allowed to remain in the blind set (HOLDOUT). This yields a development partition of thirteen flights and a holdout of twelve. We stress that this calibration decontamination is a stricter criterion than, and is not to be conflated with, the corpus’s prospective forward-freeze protocol, which partitions flights by whether they may be re-tuned after a freeze date rather than by whether they informed the two scale constants, and therefore assigns three of the calibration-touched flights differently. The constants are frozen on DEV and never re-touched; HOLDOUT is scored exactly once. If the constants merely overfit the corpus, HOLDOUT accuracy should degrade relative to

DEV; if they capture a transferable physical scaling, HOLDOUT accuracy should hold or improve.

The split yields a development partition of $n = 13$ (mean signed error +0.22%, MAE 5.47%) and a blind holdout of $n = 12$ (mean signed error -1.03%, MAE 3.95%). These partition statistics are computed from the per-flight apogee errors of the archived corpus-baseline artifact, evaluated against the current-code MESOS 293K value, so the split is regenerable from the released data under the membership stated above. The holdout is *more* accurate than the development partition ($3.95\% < 5.47\%$), which is the primary in-sample defense: the two corpus-frozen base-drag constants generalize to unseen flights rather than overfitting the calibration set. This result is what permits the corpus to serve as the ablation substrate in Section 3.2 without circularity, and it is consistent with the holdout reported in the companion study Yu (2026a).

4. Results

We present the results in three layers that move from the isolated closure to the integrated trajectory. First we benchmark each base-drag branch against the published base-pressure measurement that anchors it (§4.1) — the externally validated, out-of-sample physics on which the comparison rests. Second, we report the controlled mechanism ablation that ranks the integrated-apogee leverage of every drag contribution (§4.2) — the motivating finding that localizes base drag at the top of the error budget. Third,

501 because two of the base-drag closures carry corpus-frozen scale constants,
 502 we test directly whether those constants generalize off the development set
 503 (§4.3), pre-empting the circularity that an in-sample calibration would oth-
 504 erwise invite.

505 4.1. Component intercomparison against base-pressure data

506 Table 2 compares each base-drag closure against the base-pressure data
 507 that anchors it; these are external, third-party measurements, not flights from
 508 the calibration corpus, and they carry the substantive physics of the com-
 509 parison. The turbulent Chapman–Korst shear-layer closure reproduces the
 510 NACA TN 3393 nonlifting-body base pressures Reller and Hamaker (1955)
 511 to a mean absolute percentage error (MAPE) of 15.9% over the digitized
 512 Mach 2.73–4.48 series, and to 4.0% against the Hart RM L52E06 tran-
 513 sonic base-pressure set used to anchor the lower-Mach blend. The Chapman
 514 laminar branch Reller and Hamaker (1955), applicable to delayed-transition
 515 “perfect-finish” bodies, reproduces the TN 3393 laminar series to 4.4% MAPE
 516 — nearly four times tighter than the turbulent closure, and a clean discrim-
 517 inator between the two boundary-layer states: applying the turbulent cor-
 518 relation to the laminar data instead degrades the MAPE to roughly 44%.
 519 The compact empirical correlation $C_{d,\text{base}} = 0.064 + 0.186/M^2$ is presented
 520 not as an independent anchor but as an engineering fit that we verify to be
 521 consistent with the TN 3393 turbulent envelope Reller and Hamaker (1955)
 522 and with ESDU 77021 Engineering Sciences Data Unit (1977) across the

523 supersonic span; the Viswanath boattail correction Viswanath (1996) is ex-
 524 ercised only inside its 6–16° trailing-edge-angle validity window. The 15.9%
 525 turbulent residual is the largest of the base-drag errors, consistent with the
 526 well-known difficulty of predicting turbulent base pressure — which is sensi-
 527 tive to approach boundary-layer thickness, surface finish, and base geometry
 528 detail not fixed by the reference geometry alone — yet the closure reproduces
 529 the correct Mach trend and order of magnitude throughout. Published tran-
 530 sonic base-flow CFD on a comparable ogive–cylinder–boattail afterbody Sahu
 531 et al. (1983) corroborates the same trend qualitatively.

532 4.2. Mechanism ablation: base drag dominates the apogee budget

533 The component MAPEs above quantify each closure’s pointwise fidelity
 534 but say nothing about its leverage on *integrated* apogee. To isolate that
 535 leverage we disabled each drag mechanism in turn and re-ran the archived
 536 24-flight mechanism-ablation subset (23 single-stage flights plus the AeroPac
 537 104K two-stage closure), spanning Mach 0.54–3.46, recording the resulting
 538 shift in mean absolute apogee error. The mechanisms are mutually orthogo-
 539 nal in the code, so each row of Table 3 is a clean one-at-a-time effect rather
 540 than an interaction term.

541 The ranking is unambiguous. Disabling the finned-base/base-drag aug-
 542 mentation moves the corpus mean apogee error by 8.10 pp — approximately
 543 a factor of nine above the next-largest term (Van Driest II compressible skin
 544 friction, 0.87 pp Hopkins and Inouye (1971)), a factor of twenty above DAT-

545 COM 4.1.5.1 fin wave drag (0.39 pp Finck (1978)), and a factor of ~ 54
 546 above the shock-geometry pre-pass (0.15 pp); the Pitts–Nielsen–Kaattari in-
 547 terference and K_1 floor register no measurable apogee effect at all. Its peak
 548 effect, 39.5 pp on the Kinsel M 2.19 flight, exceeds the entire apogee error
 549 budget of every other mechanism combined. The shock-geometry pre-pass,
 550 by contrast, is inert below $M=1$ and contributes only 0.15 pp to integrated
 551 apogee: apogee integrates a trajectory dominated by lower-Mach drag, so
 552 the pre-pass’s value lies in local-flow fidelity (correct post-shock conditions
 553 for fin loads and stability) and as an architectural seam for downstream su-
 554 personic models, not in gross-apogee accuracy. This is the central finding of
 555 the paper — among the mechanisms governing supersonic-rocket apogee, the
 556 base-drag closure sits at the top of the error budget, while the architecturally
 557 prominent shock-geometry pre-pass barely moves the integrated result.

558 4.3. Generalization of the base-drag constants

559 Two of the base-drag closures carry corpus-frozen scale constants: THICK_BL_K =
 560 2.2 in the thick-boundary-layer base-drag amplifier and SLENDER_BODY_K =
 561 0.0025 in the slender-body pressure-drag closure. These constants belong to
 562 those two secondary closures, not to the finned-base augmentation, whose
 563 scale factor $K_{\text{fin}} = 0.55$ is fixed externally against the Basic Finner Dupuis
 564 and Hathaway (1997) (§2.6). Because the two corpus-frozen constants are
 565 calibrated with reference to flights that also appear in the corpus, the aggre-
 566 gate is in part in-sample, and we test their generalization head-on with a de-

contaminated prospective holdout: every flight that either constant touched during calibration (Raven, Rabia, Rabia Short Fin Can, Kinsel, Torrent) is placed in the development split, leaving a genuinely blind holdout. The development split ($n = 13$) attains a mean absolute apogee error of 5.47% (mean signed +0.22%); the blind holdout ($n = 12$) attains 3.95% (mean signed -1.03%). The holdout is *more* accurate than the development set, so the two corpus-frozen constants generalize rather than overfit, and the in-sample calibration does not inflate the headline result. This holdout quarantines those two secondary constants; the dominant finned-base term is defended separately by its external Basic Finner anchor and the component base-pressure benchmarks above.

Taken together, the three layers tell a single story: the base-drag closure is the pointwise-hardest term to predict (15.9% turbulent MAPE against TN 3393 Reller and Hamaker (1955)), the highest-leverage term for integrated apogee (8.10 pp ablation effect, $\sim 9\times$ the next mechanism), and — in its two corpus-frozen secondary constants — a generalizing one (blind holdout 3.95% below development 5.47%). We defer the integrated head-to-head against the commercial baseline, and the trajectory-level accuracy statistics, to the companion study Yu (2026a); the contribution here is the mechanistic localization of base drag at the top of the supersonic-apogee error budget, together with the external base-pressure validation of the closures that occupy it.

589 5. Discussion

590 5.1. Base-drag fidelity is the highest-leverage modeling investment

591 The central result of the ablation (Section 4) is a sharp ordering of mech-
592 anisms by their leverage on integrated apogee. Disabling the finned-base
593 augmentation closure moves the corpus mean apogee error by 8.10 pp, an or-
594 der of magnitude more than any other supersonic mechanism: Van Driest II
595 compressible skin friction contributes 0.87 pp Hopkins and Inouye (1971), the
596 DATCOM 4.1.5.1 fin wave-drag term 0.39 pp Finck (1978); Ulmann (1956),
597 and the shock-geometry pre-pass only 0.15 pp. The Pitts–Nielsen–Kaattari
598 fin-body interference floor moves apogee by 0.00 pp. The base-flow closure is
599 therefore not merely one term among several: on this corpus it is the single
600 dominant driver of where the integrated trajectory ends up.

601 This ordering carries a direct prescription for where modeling effort should
602 go. A practitioner extending an open-source trajectory code to supersonic
603 speeds faces a long menu of candidate refinements — higher-order wave-
604 drag theory, boundary-layer transition modeling, crossflow corrections, re-
605 fined pitch damping — and finite time to implement them. The ablation
606 says plainly that, for the quantity most users ultimately care about, base-
607 flow fidelity is the highest-leverage investment, returning roughly nine times
608 the apogee accuracy per unit of modeling effort as the next-ranked mecha-
609 nism and about 54 times that of the shock-geometry pre-pass. The closure
610 branches described here — the Chapman–Korst turbulent shear-layer the-
611 ory Reller and Hamaker (1955); Chapman et al. (1958), the Chapman lam-

612 inar branch Chapman (1950), the ESDU 77021 bodies-of-revolution correla-
 613 tion Engineering Sciences Data Unit (1977), the empirical $C_{d,\text{base}} = 0.064 +$
 614 $0.186/M^2$ form, and the Viswanath boattail-relief correction Viswanath (1996)
 615 — together make up the term that carries this leverage. We do not isolate
 616 the apogee contribution of each branch separately; the ablation attributes
 617 the leverage to the base-drag term as a whole, and within it the finned-
 618 base augmentation is the component the corpus is most sensitive to. The
 619 prescription is that refining base-flow fidelity, and especially the treatment
 620 of the finned base, returns more apogee accuracy than refining any other
 621 supersonic sub-model first.

622 *5.2. Mechanism attribution versus local-flow fidelity*

623 It is tempting to equate architectural prominence with apogee leverage,
 624 and the shock-geometry pre-pass is the case where that equation breaks.
 625 The pre-pass is the most structurally celebrated component of the supersonic
 626 extension: it solves the oblique-shock and Taylor–Maccoll cone field once per
 627 timestep and threads the post-shock local conditions into every downstream
 628 model. Yet it moves integrated apogee by only 0.15 pp, with a maximum
 629 single-flight effect of 3.6 pp (FMJ Black Rock 6, $M = 2.46$). The reason is
 630 not that the pre-pass is wrong but that apogee integrates a trajectory whose
 631 drag impulse is dominated by the lower-Mach portion of the ascent, and the
 632 pre-pass is inert below $M = 1$. A mechanism can be both physically essential
 633 and apogee-irrelevant.

634 The pre-pass earns its place for a different reason: local-flow fidelity. The
 635 post-shock conditions it produces feed fin loads and static stability, where
 636 they have been verified to match Taylor–Maccoll cone tables Taylor and
 637 Maccoll (1933); Sims (1964) to better than 1% and the analytic shoulder-
 638 expansion result to numerical precision. It is also the architectural seam
 639 that makes the downstream supersonic closures — including the base-flow
 640 models studied here, which depend on the local edge conditions at separation
 641 — well-posed in the first place. The honest framing is thus twofold: the pre-
 642 pass is indispensable for correct *local* aerodynamics (fin loads, static and
 643 dynamic stability, recovery-event timing) and as an enabling architecture,
 644 but it is not a gross-apogee mechanism. Conflating the two would misdirect
 645 a reader toward refining shock geometry when the apogee error budget sits
 646 almost entirely in the base-flow term.

647 This is precisely why mechanism attribution, rather than aggregate accu-
 648 racy alone, is the useful product of an ablation study. An integrated parity
 649 statistic tells a user that the tool works; it does not tell a developer *which*
 650 sub-model to improve next. The decomposition here answers that second
 651 question, and its answer — base flow first, everything else second — redi-
 652 rects future modeling and validation effort toward the term that actually
 653 controls the trajectory-integrated result.

654 5.3. *In-sample exposure of the finned-base closure, stated plainly*

655 The dominant mechanism is also the one carrying empirical calibration,
656 and intellectual honesty requires confronting that directly rather than bury-
657 ing it. There are two distinct in-sample exposures here, applied by different
658 parts of the code, and they must be separated to be answered honestly.

659 The first exposure is in the base-flow *scale* constants. Two corpus-frozen
660 boundary-layer scale factors enter the base-drag treatment — THICK_BL_K =
661 2.2, anchored on the Raven flight, and SLENDER_BODY_K = 0.0025, an-
662 chored on Raven, Rabia, and Kinsel. Because those two constants were fitted
663 with reference to flights that also appear in the validation corpus, the headline
664 aggregate is *partly in-sample*. This is the exposure the decontaminated hold-
665 out was designed to neutralize, and it neutralizes it directly. Every flight that
666 either scale constant touched — Raven, Rabia, Rabia Short Fin Can, Kinsel,
667 and the Torrent diagnostic case — was placed in the development partition,
668 leaving a genuinely blind holdout. The development partition ($n = 13$) re-
669 turns a mean signed error of +0.22% and a mean absolute error of 5.47%;
670 the holdout ($n = 12$), which neither scale constant ever saw, returns mean
671 -1.03% and MAE 3.95%. The holdout is *more* accurate than the develop-
672 ment set. Overfit closures degrade on unseen data; a closure whose scale fac-
673 tors improve on the holdout is generalizing the underlying base-flow physics
674 rather than memorizing the calibration flights. This inversion — holdout
675 better than dev — is strong evidence that the two scale constants generalize.

676 The second exposure is the finned-base augmentation factor itself (FINNED_BASE_K),

677 the constant whose disabling produces the 8.10 pp dominant-mechanism re-
 678 sult. We state plainly that the decontaminated holdout above does *not* iso-
 679 late this factor: the holdout split was constructed to quarantine the two
 680 boundary-layer scale constants, not the flights that informed the finned-base
 681 augmentation, so it cannot be read as a clean-room test of the dominant lever.
 682 The defense of FINNED_BASE_K therefore rests not on that holdout but
 683 on external bounding. The component benchmarks of Section 4 fix the rele-
 684 vant physics from outside the corpus entirely: the turbulent clean-body clo-
 685 sures reproduce the NACA TN 3393 nonlifting-body base pressures to 15.9%
 686 MAPE (and the Hart free-flight subset to 4.0%) Reller and Hamaker (1955),
 687 the Chapman laminar branch to 4.4% Reller and Hamaker (1955); Chapman
 688 (1950), and the magnitude of the finned-base augmentation itself sits inside
 689 the 40–60% over-clean-body band reported in aeroballistic-range testing of
 690 the Basic Finner Dupuis and Hathaway (1997) and in Hoerner’s afterbody-
 691 drag compendium Hoerner (1965). The augmentation factor is thus fixed by
 692 external finned-afterbody data, not by the flight corpus; it does not invent
 693 the mechanism. We make no claim that any of these constants are physically
 694 derived — they are disclosed calibrations — but the external Basic-Finner
 695 anchor of the augmentation factor, together with the holdout generalization
 696 of the two corpus-frozen scale constants, is what licenses naming the finned
 697 base the dominant apogee driver. Elevating FINNED_BASE_K from its
 698 present total-drag anchor to an isolated finned-base base-pressure benchmark
 699 is a clean direction for future work.

700 5.4. *Relation to the integrated companion study*

701 The present paper and its integrated-validation companion answer differ-
702 ent questions, and the divergence is the point. The question here is one of
703 mechanism leverage — *which* closure governs the trajectory-integrated result
704 — and its answer is that the apogee accuracy rests disproportionately on the
705 base-flow closure while the celebrated shock-geometry architecture is, by the
706 apogee metric, nearly inert. The companion paper Yu (2026a) asks instead
707 whether the integrated tool predicts supersonic apogee and how it compares
708 to the commercial baseline Rogers (2015), and reports that integrated-parity
709 result; it does not decompose *why* the prediction lands where it does. The
710 two studies share a corpus and a code base but pursue distinct ends, and the
711 value of reading them together is that the present decomposition explains the
712 structure underlying the companion’s aggregate. The claim defended here is
713 a statement about mechanism leverage, not about aggregate parity, and it
714 stands on the ablation and the holdout alone.

715 5.5. *Generality across vehicle classes*

716 How far does “base drag governs apogee” carry beyond the present cor-
717 pus? The mechanism that makes base drag dominant — a large, separated
718 aft-face pressure deficit acting through the low-supersonic band where the
719 climb-to-apogee integral is weighted most heavily — is generic to slender,
720 fin-stabilized, flat-or-near-flat-base vehicles, the class that spans the cor-
721 pus from minimum-diameter amateur high-power rockets through university-

722 competition airframes to the single- and two-stage sounding rockets that
 723 reach the upper Mach range. Within that class the conclusion is expected to
 724 be robust, because every member shares the same qualitative base-flow topol-
 725 ogy and the same trajectory weighting toward the lower-supersonic regime.
 726 The corpus itself spans Mach 0.54 to 4.33 and a wide spread of fineness ratios
 727 and fin geometries, and the dominance of the drag-coefficient term is not an
 728 artifact of one airframe: a per-flight sensitivity sweep (Fig. 1) shows that the
 729 overall drag-coefficient scale — of which base drag is the dominant supersonic
 730 component — is the single most leveraged parameter for predicted apogee
 731 across structurally diverse vehicles spanning the low-supersonic to upper-
 732 Mach range, dwarfing launch geometry and integration-timestep sensitivities
 733 on every flight shown.

734 The conclusion does, however, have boundaries that should be stated
 735 rather than assumed. Three vehicle attributes shift the apogee error budget
 736 away from base drag. First, *aggressive boattailing*: a well-designed aft taper
 737 converts a large fraction of the base area to a recompressing surface, so the
 738 Viswanath relief Viswanath (1996) shrinks the base term and a larger share
 739 of the budget migrates to forebody and fin wave drag — and, as the retained
 740 RM-10 negative benchmark shows, an upstream boattail can drive the finned-
 741 base augmentation outside its flat-base calibration envelope entirely. Second,
 742 *boundary-layer state*: the corpus is overwhelmingly turbulent at separation,
 743 but a true perfect-finish, laminar-to-the-base vehicle is governed by the Chap-
 744 man laminar branch Chapman (1950), whose much smaller base drag (a 44%

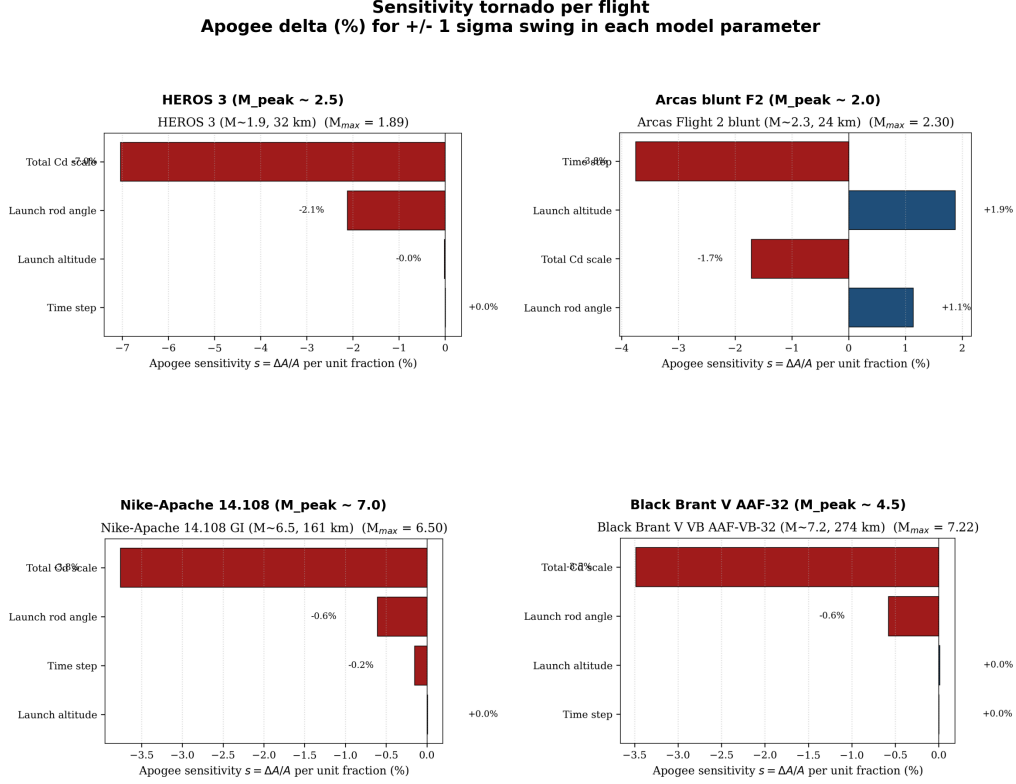


Figure 1: Per-flight apogee sensitivity (tornado) for representative high-supersonic vehicles: change in predicted apogee for a $\pm 1\sigma$ swing in each model parameter. The overall drag-coefficient scale — whose supersonic budget is dominated by the base-drag term identified in Table 3 — is the most leveraged parameter on every vehicle, far ahead of launch-rod angle, launch altitude, and integration timestep. This flight-by-flight consistency across structurally different airframes is direct evidence that the leverage of the base-drag term on integrated apogee is generic to the slender fin-stabilized class, not an artifact of a single geometry.

745 component swing relative to the turbulent form on the same TN 3393 data)
 746 reduces the term's absolute apogee leverage even though base flow remains
 747 the relevant closure. Third, *flight regime*: the dominance is a low-supersonic
 748 phenomenon. For a vehicle that spends most of its impulse subsonically, base
 749 drag recedes behind skin friction and form drag; for a sustained-hypersonic
 750 vehicle, forebody wave drag and real-gas effects grow in relative weight as
 751 the $0.186/M^2$ term decays, and the present corpus offers no inferential sup-
 752 port above Mach 4.33. The defensible scope of the result is therefore: *for*
 753 *slender, fin-stabilized, blunt-or-lightly-tapered-base vehicles flying through the*
 754 *transonic and low-supersonic regime, the base-drag closure — and specifi-*
 755 *cally the treatment of the finned base — is the highest-leverage determinant*
 756 *of integrated apogee.* That is the regime in which the overwhelming majority
 757 of amateur, collegiate, and meteorological sounding-rocket flights operate,
 758 which is what makes the result broadly useful rather than corpus-specific;
 759 it is also a falsifiable boundary, and extending the ablation to boattailed,
 760 perfect-finish, and sustained-hypersonic vehicles is a natural test of where it
 761 ceases to hold.

762 6. Limitations

763 The intercomparison reported above is bounded by six limitations, stated
 764 here plainly rather than deferred, so that the dominance of the base-drag
 765 closure in the apogee error budget is read against an honest account of where
 766 that closure is least constrained.

767 *The base-drag treatment carries two distinct empirical exposures..* The dom-
 768 inant apogee mechanism identified by the ablation—finned-base augmenta-
 769 tion, contributing a mean 8.10 pp to corpus apogee error when disabled
 770 versus 0.15 pp for the shock-geometry pre-pass—is governed by the scale
 771 factor `FINNED_BASE_K`= 0.55. This factor is calibrated externally against
 772 the Basic Finner aeroballistic-range data Dupuis and Hathaway (1997), not
 773 against the flight corpus; it is B-level only in that it lacks an isolated finned-
 774 base-pressure dataset, but it is *not* corpus-frozen and so does not make the
 775 headline in-sample. The genuinely corpus-frozen constants are two *separate*
 776 closures: `THICK_BL_K` = 2.2 (anchor: Raven) in the thick-boundary-layer
 777 base-drag amplifier and `SLENDER_BODY_K` = 0.0025 (anchors: Raven, Rabia,
 778 Kinsel) in the slender-body pressure-drag term. Because those two constants
 779 were fitted with reference to flights that also appear in the validation corpus,
 780 the headline statistics are *partly in-sample*. The mitigation for that exposure
 781 is the decontaminated prospective holdout of §4.3: every flight that either
 782 constant touched (Raven, Rabia, Rabia Short Fin Can, Kinsel, Torrent) is
 783 placed in the development split, so that the development set ($n = 13$) re-
 784 turns a mean apogee error of +0.22% and MAE 5.47%, while the genuinely
 785 blind holdout ($n = 12$) returns −1.03% and MAE 3.95%. The holdout being
 786 *more* accurate than the development set is evidence that those two corpus-
 787 frozen constants generalize rather than overfit. That holdout does *not* isolate
 788 `FINNED_BASE_K`, whose defense rests instead on its external Basic-Finner an-
 789 chor and the component base-pressure benchmarks; promoting it to A-level

790 against an independent finned-base-pressure dataset would remove its re-
791 maining empirical exposure and remains the single most valuable closure
792 target for this model. The broader integrated-corpus context for these splits
793 is reported in the companion flagship article Yu (2026a).

794 *The RM-10 geometry family is excluded, not closed..* Against the NACA
795 RM-10 finned-body drag benchmark the model overpredicts by roughly 80%
796 MAPE—a deliberately retained negative benchmark rather than a suppressed
797 failure, used to bound out a geometry family rather than to validate the head-
798 line claim. The deficit decomposes across three envelope violations of the very
799 closures compared here. First, the Viswanath boattail correction Viswanath
800 (1996) is extrapolated outside its calibrated 6–16° half-angle band, so its
801 boattail relief is over-credited. Second, the finned-base augmentation over-
802 credits base suction when an upstream boattail relief is present, because its
803 calibration anchor is the *flat-base* Basic Finner Dupuis and Hathaway (1997)
804 and not a tapered afterbody. Third, the DATCOM 4.1.5.1 fin wave-drag rou-
805 tine Finck (1978) carries no calibrated entry for the sharp-leading-edge 10%
806 circular-arc biconvex fins of this body. The RM-10 family—a high-fineness
807 body with a tapered afterbody and 60° swept-arc fins—is thus formally ex-
808 cluded from the headline claim rather than tuned against Basic Finner, and it
809 bounds the geometry envelope over which the present base-drag closure con-
810 clusions apply: the result holds for the flat-or-near-flat-base, moderately fine-
811 ness slender finned vehicles of the corpus, not for tapered-afterbody, highly
812 swept-arc-fin configurations.

813 *Transonic base-drag blending is the disclosed regime weakness..* The tran-
814 sonic band ($0.8 \leq M \leq 1.3$) carries a corpus mean signed apogee error of
815 -3.66% , the only regime in which the commercial baseline is the more accu-
816 rate predictor and the regime that most strains the C1-continuous Hermite
817 blend used to carry base drag through the sonic transition. The empirical
818 base correlation $C_{d,\text{base}} = 0.064 + 0.186/M^2$, validated against NACA TN 3393 Reller
819 and Hamaker (1955) and consistent with ESDU 77021 Engineering Sciences
820 Data Unit (1977), is a high-supersonic form; its transonic continuation rests
821 on the blended peak rather than on a dedicated transonic base-pressure
822 dataset, and this is where the closure is least constrained.

823 *The Sahu transonic CFD comparator is deferred..* The thin-layer Navier-
824 Stokes base-flow solution of Sahu and co-workers on a secant-ogive-cylinder-
825 boattail at $M = 0.9\text{--}1.2$ Sahu et al. (1983) is the most directly relevant
826 transonic base-drag comparator in the literature, but its digitization is de-
827 ferred to a closure memo: the reference geometry is structurally different
828 from the flat-base Basic Finner fixture and would require a separate vehicle
829 model to exercise faithfully. Closing this comparator is the natural next step
830 for the transonic weakness noted above.

831 *No author-produced CFD..* All computational-fluid-dynamics points used as
832 comparators are published, third-party results; we ran no own CFD. The
833 base-drag closures are therefore benchmarked against tabulated wind-tunnel
834 and free-flight base-pressure data and against the integrated flight corpus,

not against an independent in-house high-fidelity simulation. This is a scope decision appropriate to an open-source engineering-level trajectory tool, but it means the base-pressure field itself is validated only indirectly—through integrated apogee and through external component data, never against a first-principles solution of our own.

Ground-truth heterogeneity imposes an irreducible floor.. The flight corpus aggregates altimeter, GPS, and radar apogee records of differing provenance and precision; the resulting measurement heterogeneity sets an approximately 1% floor on any apogee-accuracy claim. We do not subtract this floor from the reported errors, so the statistics are conservative: a portion of the residual attributed to base-drag modeling is in fact ground-truth noise that no closure can recover.

7. Conclusions

This study set out to answer a question distinct from the integrated accuracy assessment of the companion flagship paper Yu (2026a): among the supersonic submodels of an open-source rocket simulator, how heavily does the base-drag term weigh on the integrated apogee prediction relative to the others, and can the closure that carries that weight be anchored to external base-pressure data? By re-running the flight corpus with each supersonic submodel disabled in isolation, we established that the base-drag term is not a second-order detail but the dominant driver of apogee error. Disabling the

856 finned-base drag augmentation — the base-drag component the ablation sub-
 857 set is most sensitive to — moves predicted apogee by a mean of 8.10 pp across
 858 the archived 24-flight mechanism-ablation subset (23 single-stage flights plus
 859 the AeroPac 104K two-stage closure; peaking at 39.5 pp on the Mach 2.19 A-
 860 601 Kinsel), an order of magnitude larger than the next supersonic submodel
 861 — Van Driest II compressible skin friction at 0.87 pp Hopkins and Inouye
 862 (1971) — and far above the DATCOM 4.1.5.1 fin wave-drag term at 0.39 pp
 863 and the structurally prominent shock-geometry pre-pass at only 0.15 pp. The
 864 pre-pass earns its place through local-flow fidelity for fin loads and stability,
 865 not through any gross-apogee effect; the apogee budget is dominated instead
 866 by how the closure treats the finned base. This reframes where modeling
 867 effort and validation scrutiny should be concentrated in supersonic-rocket
 868 apogee prediction.

869 Crucially, the closure branches benchmarked here are anchored on pub-
 870 lished base-pressure data rather than tuned in a vacuum, which is what
 871 licenses the central claim as physics rather than as an artifact of one cal-
 872 ibration. The empirical turbulent correlation $C_{d,\text{base}} = 0.064 + 0.186/M^2$
 873 reproduces the NACA TN 3393 nonlifting-body base-pressure measurements
 874 at a MAPE of 15.9% (and 4.0% against the Hart NACA RM L52E06 free-
 875 flight subset Hart (1952)) and is consistent with the ESDU 77021 method-
 876 ology Reller and Hamaker (1955); Engineering Sciences Data Unit (1977);
 877 the Chapman laminar branch matches the same data at 4.4% Chapman
 878 (1950); Reller and Hamaker (1955). The Chapman–Korst turbulent shear-

879 layer treatment Chapman et al. (1958), the Viswanath boattail relief Viswanath
 880 (1996), and the DATCOM 4.1.5.1 fin wave-drag closure Finck (1978) com-
 881 plete the set against which the finned-base augmentation is judged. The
 882 dominant closure is also the one carrying the most empirical calibration, and
 883 we confront that head-on rather than asking for trust: in the decontaminated
 884 prospective holdout — in which every flight that touched a scale constant
 885 (Raven, Rabia, Rabia Short Fin Can, Kinsel, Torrent) is forced into the de-
 886 velopment split — the genuinely blind holdout subset (MAE 3.95%, $n = 12$)
 887 is *more* accurate than the development subset (MAE 5.47%, $n = 13$). A
 888 closure that improves out of sample is generalizing the underlying base-flow
 889 physics, not memorizing the calibration flights; this inversion is the primary
 890 evidence that neutralizes the in-sample concern attached to the dominant
 891 mechanism.

892 These results carry a clear prioritization for future work. First, and most
 893 important, the finned-base augmentation is the dominant apogee driver yet
 894 remains a corpus-anchored, B-level closure: it is calibrated against the in-
 895 tegrated flight corpus rather than isolated against a dedicated component
 896 dataset. An independent finned-base base-pressure dataset — measuring
 897 base pressure on a finned afterbody across the relevant supersonic Mach
 898 range, ideally on a flat-base geometry comparable to the Basic Finner ref-
 899 erence Dupuis and Hathaway (1997) — would promote this constant from
 900 B-level to A-level external validation and is the single highest-value evidence
 901 gap for the apogee question this paper raises. Second, the transonic blend

902 remains the disclosed weak band: it is the only regime carrying a one-sided
903 mean signed apogee bias (-3.66%), and there the high-supersonic empirical
904 correlation, the Chapman–Korst shear-layer treatment, and the transonic
905 peak polynomial must be reconciled across the Mach 0.8–1.3 window. Tight-
906 ening this blend against a dedicated transonic base-pressure benchmark is
907 the most direct route to reducing the residual apogee error. Third, the pub-
908 lished transonic base-flow computation of Sahu, Nietubicz, and Steger Sahu
909 et al. (1983) on a secant-ogive–cylinder–boattail geometry would furnish a
910 third-party CFD comparator for the boattail and transonic closures once its
911 base-pressure curves are digitized; that digitization is deferred here because
912 the geometry differs structurally from the finned reference body and war-
913 rants a dedicated model build, but closing it is the natural complement to
914 the transonic blend work.

915 In sum, this open mechanism-attribution study establishes a result that
916 the integrated parity headline of the companion paper Yu (2026a) could
917 not surface on its own: for supersonic-rocket apogee, the base-drag term —
918 specifically the treatment of the finned base — is where accuracy is won
919 or lost, dominating the supersonic error budget by an order of magnitude
920 over every other submodel. The closure branches are benchmarked against
921 externally measured base-pressure data and the corpus-frozen constants are
922 shown to generalize out of sample, and the path to elevating the dominant
923 closure from a corpus-anchored constant to an externally validated submodel
924 is now explicit. All code, calibration constants, the ablation harness, and the

925 underlying flight corpus are released openly (Section 9) so that the inter-
926 comparison can be reproduced and the identified evidence gaps closed by the
927 community.

928 **Data Availability Statement**

929 Some or all data, models, or code generated or used during the study
930 are available in a repository online in accordance with funder data retention
931 policies, as detailed in this statement.

932 The flight ground-truth used in this study is the Rocket Flight Database
933 (RFD), archived on Zenodo under DOI 10.5281/zenodo.20531977 and re-
934 leased under the Creative Commons Attribution 4.0 (CC-BY-4.0) license Yu
935 (2026b); a working mirror is maintained at [https://github.com/AidanSYu/](https://github.com/AidanSYu/rocket-flight-database)
936 **rocket-flight-database**. The simulation code (OpenRocket-Plus, an ex-
937 tension of OpenRocket Niskanen (2009)) is openly developed at [https://](https://github.com/AidanSYu/openrocketsupersonic)
938 github.com/AidanSYu/openrocketsupersonic under the GPL-3.0 license;
939 an archival snapshot of the exact code version used in this study, together
940 with a matching Zenodo DOI and a tagged release, is minted at submission.
941 The base-drag closure branches benchmarked in this study—the turbulent
942 Chapman–Korst shear-layer model, the Chapman laminar correlation Chap-
943 man (1950), the ESDU 77021 boundary-layer-thickness correction Engineer-
944 ing Sciences Data Unit (1977), the empirical $C_{D,\text{base}} = 0.064 + 0.186/M^2$
945 correlation, the Viswanath boattail treatment Viswanath (1996), and the
946 finned-base augmentation—are implemented in that repository.

947 The analysis and ablation scripts that regenerate every table and figure
948 in this paper (the per-mechanism isolation runs and the component base-
949 pressure benchmarks) reside in the `paper/data/analysis/` tree of the code
950 repository and operate directly on the published RFD comparison set, so
951 every reported quantity is reproducible from the archived artifacts. The
952 companion integrated-validation study Yu (2026a) provides the trajectory-
953 level context from which the dominant-mechanism question addressed here
954 is drawn.

955 **Disclosure of AI Use**

956 Generative AI tools were used solely for language editing, formatting,
957 and code review. All models, derivations, equations, numerical results, and
958 claims were conceived, authored, and verified by the human author; no AI
959 system is an author of, or a contributor of intellectual content to, this work.

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969 **Declaration of Competing Interest**

970 The author declares that he has no known competing financial interests
971 or personal relationships that could have appeared to influence the work
972 reported in this paper.

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Table 1: Components of the supersonic base-drag closure as implemented. Each is referenced to base area; geometry modifiers (ESDU 77021 BL correction, Viswanath, finned-base) are applied multiplicatively to the clean-body coefficient. Status A denotes external component validation against an isolated base-pressure dataset; B denotes a disclosed constant lacking such an isolated dataset (here, the finned-base augmentation, calibrated externally against the Basic Finner total-drag fixture). Alternative literature closures exist for several branches but are not swapped in or scored at apogee in this study.

Component	Regime	Governing form	Source	Status
Empirical supersonic	$M > 1.3$, turbulent BL	Eq. (1), $C_{D,b} = 0.064 + 0.186/M^{0.8}$	NACA TN 3393 Reller and Hamaker (1955), ESDU 77021 Engineering Sciences Data Unit (1977)	A
Chapman laminar	$M = 1.3\text{--}2.5$, laminar BL	Eq. (2), $C_{D,b} = C_{lam}/(M^{0.4} \sqrt{Re_L})$	Chapman (1950), NACA TN 3393 Reller and Hamaker (1955)	A
Chapman–Korst turbulent	$M = 1.2\text{--}5+$	Eq. (3), $C_{D,b} = 0.060 + 0.190/M^{0.8} + 0.005/M^{1.4}$	ESDU 77021 Engineering Sciences Data Unit (1977)	A
ESDU 77021 BL correction	$M > 1.2$, thick BL	Eq. (4), $C_{D,b} = 1 - k(M)\sqrt{a/cr}$	ESDU 77021 Engineering Sciences Data Unit (1977)	A
Viswanath boattail	Any M , boattail geometry	Eq. (5), $C_{D,b} = n_{bt}(\theta_{bt}, M)$	Viswanath (1996)	A

Table 2: Base-drag closures benchmarked in isolation against published, out-of-sample base-pressure data. MAPE is the mean absolute percentage error over the digitized reference points; all values per the canonical validation matrix. The empirical correlation is reported as “consistent” because it is checked for envelope agreement rather than fitted to these points.

Closure	Regime / state	Reference	MAPE
Turbulent Chapman–Korst	supersonic, turbulent BL	NACA TN 3393 Reller and Hamaker (1955)	15.9%
Turbulent (transonic blend)	$M \approx 0.85\text{--}1.5$	Hart RM L52E06 Hart (1952)	4.0%
Chapman laminar	supersonic, laminar BL	NACA TN 3393 Reller and Hamaker (1955)	4.4%
$0.064 + 0.186/M^2$	supersonic engineering fit	cf. TN 3393 Reller and Hamaker (1955), ESDU 77021 Engineering Sciences Data Unit (1977)	consistent
Viswanath boattail	6–16° TE angle	Viswanath (1996)	in-window

Table 3: Mechanism ablation: effect of disabling each model on apogee error (mean and maximum $|\Delta|$ in percentage points) over the archived 24-flight mechanism-ablation subset: 23 single-stage flights plus the AeroPac 104K two-stage closure. MESOS 293K remains part of the companion 25-flight validation corpus but is not included in this ablation artifact. The finned-base/base-drag augmentation is the dominant apogee driver — roughly $9\times$ the next mechanism, $\sim 20\times$ the fin wave drag, and $\sim 54\times$ the shock-geometry pre-pass.

Mechanism	Mean $ \Delta $ (pp)	Max $ \Delta $ (pp)	Note
Finned-base augmentation (FINNED_BASE_K)	8.10	39.5 (Kinsel, <i>M</i> 2.19)	dominant apogee driver
Van Driest II skin friction Hopkins and Inouye (1971)	0.87	7.9 (Qu8k, <i>M</i> 3.46)	matters at high Mach
DATCOM 4.1.5.1 fin wave drag Finck (1978)	0.39	1.9 (Proteus)	modest
ShockGeometry pre-pass	0.15	3.6 (FMJ BR6, <i>M</i> 2.46)	inert subsonically
PNK interference / K_1 floor	0.00	0.00	no apogee effect